🎓 GENAI MASTERCLASS 2025

From Zero to GenAI *Hero*

LLMs · RAG · Vector DBs · Agentic AI · Chatbots · Market Leaders — Everything in 1 Week

1hr CRASH SESSION

7 DAY CURRICULUM

50+ CORE CONCEPTS

10+ MARKET LEADERS

Why GenAI? | 1-Hour Session | Core Terms | LLM Types | RAG Deep Dive | Vector DB | Agentic AI | Chatbots | Market Compare | Week Plan

// MODULE 0 — FOUNDATION

Why GenAI? The Big Picture

> *"We are at an inflection point. AI is not just a tool — it's a new way of building intelligence into everything."*

🌍

Why GenAI Exists

Traditional AI recognized patterns. GenAI *creates* — text, code, images, audio. It's a generative leap from discriminative models.

💬

Why LLMs?

Language is the universal interface. LLMs understand and generate human language at superhuman scale — 100B+ parameter reasoning engines.

📚

Why RAG?

LLMs have outdated, hallucinated knowledge. RAG connects them to real, current, domain-specific data — grounding responses in truth.

🤖

Why Chatbots?

24/7 availability, instant response, infinite scale. Replaces call centers, powers customer service, education, healthcare, banking.

🧠

Why Agentic AI?

Moving beyond Q&A. Agents take multi-step actions, use tools, browse web, write & run code, manage workflows autonomously.

📦

Why Vector DBs?

Traditional databases can't handle "meaning." Vector DBs store semantic embeddings — enabling similarity search at billion-doc scale.

💰

Market Reality

$1.3 trillion GenAI market by 2032. Every company is hiring AI engineers. Not learning this = career risk.

⚡

Speed of Change

GPT-3 (2020) → GPT-4 → Claude → Gemini → Llama 3 → GPT-o1. The pace is unprecedented. You need a solid foundation now.

// MODULE 1 — 1-HOUR CRASH COURSE

The 60-Minute GenAI Sprint

This is your live session plan. Each block is self-contained. Cover these in order for maximum impact with a new audience.

⏱ MIN 0-8 | HOOK

🚀 "AI Wrote This Slide Deck" — Opening Demo

→ Live ChatGPT/Claude demo: generate a cover letter, debug code, explain a concept
→ Show before/after: Google search vs. AI response quality
→ State the thesis: "GenAI is a platform shift like the internet"
→ Poll the room: Who has used ChatGPT? Who has built with an API?

⏱ MIN 8-18 | FUNDAMENTALS

🧠 What is a Large Language Model?

→ The transformer architecture in 2 minutes (attention = "reading between the lines")
→ Training: next-token prediction on internet-scale text

→ RLHF: How ChatGPT learned to be helpful vs. raw GPT
→ Tokens, context windows, temperature — the 3 dials that matter
→ Analogy: LLM = A brain trained on ALL books ever written, compressed

⏱ MIN 18-26 | THE PROBLEM

⚠️ Why LLMs Alone Are Not Enough — Hallucination Problem

→ Demo: Ask ChatGPT about your company. It will hallucinate.
→ Knowledge cutoff problem: training data is frozen in time
→ No access to private data (your PDFs, databases, internal docs)
→ Enter RAG: Retrieval Augmented Generation — the fix

⏱ MIN 26-38 | THE SOLUTION

📚 RAG + Vector Databases Explained

→ RAG Flow: User Query → Embed → Vector Search → Retrieve Docs → Inject into Prompt → LLM → Answer
→ What is an embedding? "Turning meaning into coordinates in space"
→ Vector DB = library where books are stored by meaning, not by name
→ Live example: Building a chatbot that answers from YOUR documents
→ Tools: Pinecone, Weaviate, ChromaDB, pgvector

⏱ MIN 38-48 | THE FUTURE

🤖 Agentic AI — Beyond Chat

→ Chat vs. Agent: Chat answers questions. Agents DO things.
→ ReAct pattern: Reason → Act → Observe → Repeat
→ Tools agents can use: Web search, code execution, email, APIs, databases
→ Multi-agent systems: Orchestrator → Sub-agents (like a team)
→ Real demo: AutoGPT / CrewAI / LangGraph running a task end-to-end

⏱ MIN 48-56 | MARKET

📊 Who's Winning? — Market Players & Landscape

→ OpenAI (GPT-4o, o1) vs. Anthropic (Claude 3.5) vs. Google (Gemini) vs. Meta (Llama)

→ Open source vs. Closed source — the debate
→ Cloud providers: AWS Bedrock, Azure OpenAI, Google Vertex AI
→ Where the money is: $10B+ raised in GenAI in 2024

⏱ MIN 56-60 | CLOSE

🗺️ Your 1-Week Roadmap + Q&A

→ Preview the 7-day curriculum
→ Key takeaway: "You now understand what 80% of AI engineers know conceptually"
→ 3 things to do tonight: Sign up for OpenAI API, clone a LangChain repo, read one RAG paper
→ Open Q&A — collect real questions to address in the week-long course

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// MODULE 2 — TERMINOLOGY

Essential **Glossary** — 30 Terms You Must Know

🔤 Token

The atomic unit of text for LLMs. "ChatGPT" = 2 tokens. ~1 word = ~1.3 tokens. Models have a max token limit (context window).

🔥 Real Use: GPT-4o supports 128K tokens — that's ~96K words or an entire novel.

👻 Hallucination

LLMs generating confident, plausible-sounding but factually wrong information. A fundamental limitation of generative models.

🔥 Real Use: A lawyer used ChatGPT to find case precedents — it fabricated 6 fake court cases.

📐 Embedding

A fixed-length array of numbers (vector) representing the semantic

🧠 Attention Mechanism

meaning of text. Similar meanings → nearby vectors in high-dimensional space.

🔥 Real Use: Google uses embeddings to power semantic search — "jaguar speed" returns car AND animal results.

🌡️ Temperature

Controls randomness in outputs. 0 = deterministic/focused, 1 = creative/random. Set to 0 for code gen, 0.8 for creative writing.

🔥 Real Use: Customer service bots use temp=0. AI storytellers use temp=0.9.

Context Window

Maximum tokens an LLM can "see" at once — its working memory. Beyond this, it forgets earlier context.

🔥 Real Use: Gemini 1.5 Pro: 1M token context = analyzing 700 PDFs at once.

🎯 Prompt Engineering

The art of crafting inputs to get optimal outputs from LLMs. Includes few-shot, zero-shot, chain-of-thought, system prompts.

🔥 Real Use: Adding "Let's think step by step" improves math accuracy by 40-80%.

🔗 Chain of Thought (CoT)

Prompting LLMs to show reasoning steps before answering. Dramatically improves accuracy on complex problems.

Core transformer innovation. Allows each token to "attend to" and weigh the relevance of all other tokens in context. Enables understanding long-range dependencies.

🔥 Real Use: Powers translation, summarization, code generation in all modern LLMs.

🔧 Tool Calling / Function Calling

LLM can decide to call external functions/APIs to get data. Bridges language models with real-world systems.

🔥 Real Use: ChatGPT searches the web, runs Python, queries databases using tool calling.

📋 System Prompt

Hidden instructions given to an LLM before the user conversation starts. Defines the AI's persona, rules, and context.

🔥 Real Use: "You are a helpful customer support agent for Zomato. Only answer food delivery questions."

🔄 RAG — Retrieval Augmented Generation

Architecture that retrieves relevant external documents at query time and injects them into the LLM prompt to ground responses in facts.

🔥 Real Use: Microsoft Copilot uses RAG over your email/docs/calendar to give personalized answers.

🔥 Real Use: Used in medical diagnosis AI to show reasoning transparency.

⚙️ Fine-tuning

Further training a pre-trained LLM on domain-specific data to specialize its behavior without changing its full weights.

🔥 Real Use: Bloomberg GPT fine-tuned on financial data. Med-PaLM fine-tuned on medical literature.

🧩 RLHF

Reinforcement Learning from Human Feedback. How ChatGPT was made helpful — humans rated outputs, reward model trained, policy updated.

🔥 Real Use: Used by OpenAI, Anthropic, Google to align models with human preferences.

🕸️ Knowledge Graph

A graph database of entities and their relationships. Used in advanced RAG (GraphRAG) to traverse structured knowledge.

🔥 Real Use: Microsoft's GraphRAG used to analyze SEC filings — finding non-obvious company relationships.

🎛️ Inference

Running a trained model to generate outputs. Distinct from training. Inference latency and cost are key production metrics.

🔥 Real Use: GPT-4 inference costs ~$10/1M tokens. Llama 3 self-hosted: ~$0.20/1M tokens.

🔡 Tokenizer

Converts raw text into tokens the model understands. Different models use different tokenizers (BPE, WordPiece). Affects cost and context use.

🔥 Real Use: Hindi/Chinese text uses more tokens per word than English — higher cost for non-English queries.

// MODULE 3 — LARGE LANGUAGE MODELS

Types of LLMs — Complete Taxonomy

Think of LLMs like cars. **Base models** are the engine. **Instruction-tuned models** are street cars. **RLHF-tuned models** are cars with a polite autopilot. **Fine-tuned models** are race cars built for one specific track.

LLM ARCHITECTURE FLOW

Raw Text (Internet) → Tokenize → Transformer Layers → Self-Attention → Pre-trained Weights → RLHF / Fine-tune → Chat Model

// MODULE 4 — RETRIEVAL AUGMENTED GENERATION

RAG Deep Dive — Types & Architecture

User Query → Embed Query → Vector Search → Top-K Docs → Build Prompt → LLM Generate → Grounded Answer

RAG Chunking Strategies

// MODULE 5 — VECTOR DATABASES

Vector **Databases** — The Memory of AI

Traditional DB: "Find all documents titled 'contract'." Vector DB: **"Find all documents that MEAN the same thing as 'legal agreement'"** — even if the word 'contract' never appears.

📐 Vector / Embedding

A list of ~1536 numbers representing the semantic meaning of a piece of text. Generated by embedding models (OpenAI, Cohere, Google).

🔥 Real Use: "King - Man + Woman = Queen" — vectors capture meaning algebraically.

🔍 ANN — Approximate Nearest Neighbor

Finding vectors most similar to a query vector. Uses algorithms like HNSW, IVF, PQ for billion-scale search in milliseconds.

🔥 Real Use: Spotify's song recommendations — find 10 similar songs

🗂️ Index

Data structure for fast vector search. HNSW (best quality+speed), IVF (memory efficient), FLAT (brute-force, exact but slow).

🔥 Real Use: Pinecone's cloud HNSW index serves 200ms latency at 1B vector scale.

🏷️ Metadata Filtering

Filter vector search by metadata fields (date, author, category) BEFORE or AFTER semantic search. Critical for multi-tenant RAG.

🔥 Real Use: "Find similar docs to this query, BUT only from last 30 days, AND in Hindi."

from 100M catalog in <10ms.

📊 Cosine Similarity

Measures angle between vectors. Value: -1 to 1. Score of 0.95+ = very similar meaning. Most common similarity metric in RAG.

🔥 Real Use: Semantic search, duplicate detection, recommendation systems.

🔀 Hybrid Search

Combine vector (semantic) + keyword (BM25/TF-IDF) search with reciprocal rank fusion. Better than either alone.

🔥 Real Use: Enterprise search engines, e-commerce product search, legal discovery.

Vector DB Comparison Table

Database	Type	Best For	Scale	Unique Strength
Pinecone ★	Managed Cloud	Production RAG	Billions	Zero infra management, fastest setup
Weaviate	Open Source / Cloud	Multi-modal RAG	100M+	Built-in vectorization, GraphQL API
Qdrant	Open Source / Cloud	Performance-critical	100M+	Rust-built, fastest open-source, filtering
Chroma	Open Source	Local dev / prototyping	Millions	Easiest to start, Python-native
Milvus	Open Source / Cloud	Enterprise on-prem	Billions	Most mature open-source, GPU support
pgvector	Postgres Extension	Existing Postgres users	Millions	No new infra, SQL-native vector search
Redis VSS	In-memory	Low-latency search	Millions	Sub-millisecond, existing Redis stack

Database	Type	Best For	Scale	Unique Strength
Elastic	Hybrid (text+vector)	Enterprise search	Billions	Best hybrid search, existing ES users

// MODULE 6 — AGENTIC AI

Agentic AI — The Autonomous Revolution

🤖 What is an Agent?

An LLM + tools + memory + ability to take action. It plans multi-step tasks, uses tools, observes results, and adapts — without human input per step.

🔥 Real Use: An agent that receives "book cheapest flight to Mumbai next week" and handles searching, comparing, booking all autonomously.

🔄 ReAct Pattern

Re

ason +

Act

. The agent thinks out loud (Thought), takes an action (Action), observes the result (Observation), repeats until done.

🔥 Real Use: Foundation of LangChain agents, OpenAI Assistants API, Claude tool use.

🧠 Agent Memory Types

Short-term:

Conversation history in context.

Long-term:

External DB (vector/SQL).

Episodic:

Past task logs.

Semantic:

General facts.

🔥 Real Use: MemGPT project: agents with infinite memory via hierarchical memory management.

👥 Multi-Agent Systems

Multiple specialized agents collaborating. Orchestrator agent directs sub-agents (researcher, writer, coder, critic). Enables complex workflows.

🛠️ Tool Use

Agents access external tools: web search, code execution, email, calendar, databases, APIs. Transforms LLMs from knowledge stores into action engines.

🔥 Real Use: AutoGPT, Claude Computer Use, OpenAI Code Interpreter.

🔥 Real Use: CrewAI: Research agent + Writer agent + Editor agent producing full blog posts autonomously.

🗺️ Planning & Reasoning

Tree of Thoughts:

Explore multiple reasoning paths.

MCTS:

Monte Carlo Tree Search for complex decisions.

Chain of Thought:

Linear step-by-step.

🔥 Real Use: OpenAI o1 uses internal CoT to solve competition math problems at expert level.

Agentic AI Frameworks

// MODULE 7 — AI CHATBOTS

Chatbots — Types & Architecture

Chatbot Stack (Modern Architecture)

User Input (Web/App/Voice) → Guard Rails (Safety Check) → Intent Router → RAG Retrieval →
LLM Generate → Output Filter → Response + Logging

// MODULE 8 — MARKET LANDSCAPE

Market **Legends** — 2025 Comparison

Company / Model	Top Model	Context	Strengths	Weakness	Pricing (input/1M)
OpenAI ★	GPT-4o, o1, o3	128K	Best ecosystem, tools, API, plugins	Most expensive, closed	$5-$15
Anthropic	Claude 3.5 Sonnet/Opus	200K	Safety, long context, coding	No image gen, newer	$3-$15
Google DeepMind	Gemini 1.5 Pro/Ultra	1M tokens	Longest context,	API inconsistency	$3.50-$10

Company / Model	Top Model	Context	Strengths	Weakness	Pricing (input/1M)
			multimodal, Search		
Meta AI ★	Llama 3.1 405B	128K	Free, open source, self-hostable	Needs infra to run	Free (self-host)
Mistral AI	Mixtral 8x22B, Mistral Large	32K	Efficient, European, open weights	Smaller ecosystem	\$2-\$6
Microsoft	Azure OpenAI + Phi-3	128K	Enterprise trust, compliance, Teams	Azure lock-in	\$1-\$30
Amazon AWS	Bedrock (multi-model)	Varies	Model choice, AWS integration	No frontier model	Varies
Cohere	Command R+	128K	RAG-optimized, enterprise	Not consumer-focused	\$0.50-\$3
DeepSeek	R1, V2.5	128K	o1-level reasoning, \$0.14/1M	China-based, privacy concerns	\$0.14-\$0.55
xAI (Elon)	Grok 2	128K	Real-time X/Twitter data, uncensored	X platform dependency	\$5

RAG / AI App Framework Leaders

Tool	Category	GitHub Stars	Best Use Case	Learning Curve
LangChain ★	LLM Orchestration	90K+	General RAG, agents, chains	Medium
LlamaIndex	Data + RAG Framework	35K+	Document Q&A, enterprise RAG	Medium
LangGraph	Multi-agent orchestration	7K+	Complex stateful agents	High
CrewAI	Multi-agent collaboration	20K+	Role-based agent teams	Low-Medium
Haystack	Production RAG	14K+	Enterprise search, pipelines	Medium
Flowise	No-code LLM builder	30K+	Visual RAG app builder	Low
Vercel AI SDK	Frontend AI	10K+	React/Next.js AI apps	Low
DSPy	Prompt optimization	17K+	Automatic prompt engineering	High

Real-World Industry Use Cases

// MODULE 9 — CURRICULUM

Your 7-Day GenAI Mastery Plan

3-4 hours/day. Mix of theory, coding, and building. By Day 7 you'll have a working RAG chatbot deployed.

📦 Tech Stack for the Week

GenAI Masterclass

1 Hour Session + 7 Day Curriculum · LLMs · RAG · Vector DB · Agentic AI · Chatbots · Market

Covers: OpenAI · Anthropic · Google · Meta · Mistral · DeepSeek · Cohere · LangChain · LlamaIndex · Pinecone · CrewAI

Batch 2025 · GenAI Fundamentals to Production